

Physiology of The Immune System



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Introduction

1. The ability to ward off disease is called immunity (resistance). Lack of resistance is called susceptibility.
2. The two general types of immunity are (a) innate and (b) adaptive (acquired).
3. Innate immunity refers to a wide variety of body responses to a wide range of pathogens.
4. Adaptive immunity involves activation of specific lymphocytes to combat a particular foreign substance.

IMMUNITY

INNATE IMMUNITY

Physical
Barriers



Chemical
Barriers

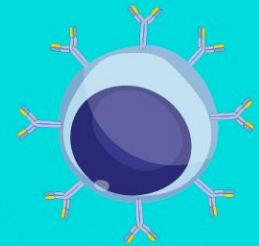


Cellular
Defences



ADAPTIVE IMMUNITY

Active
Immunity



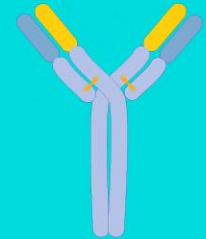
Natural



Vaccination



Passive
Immunity



Maternal



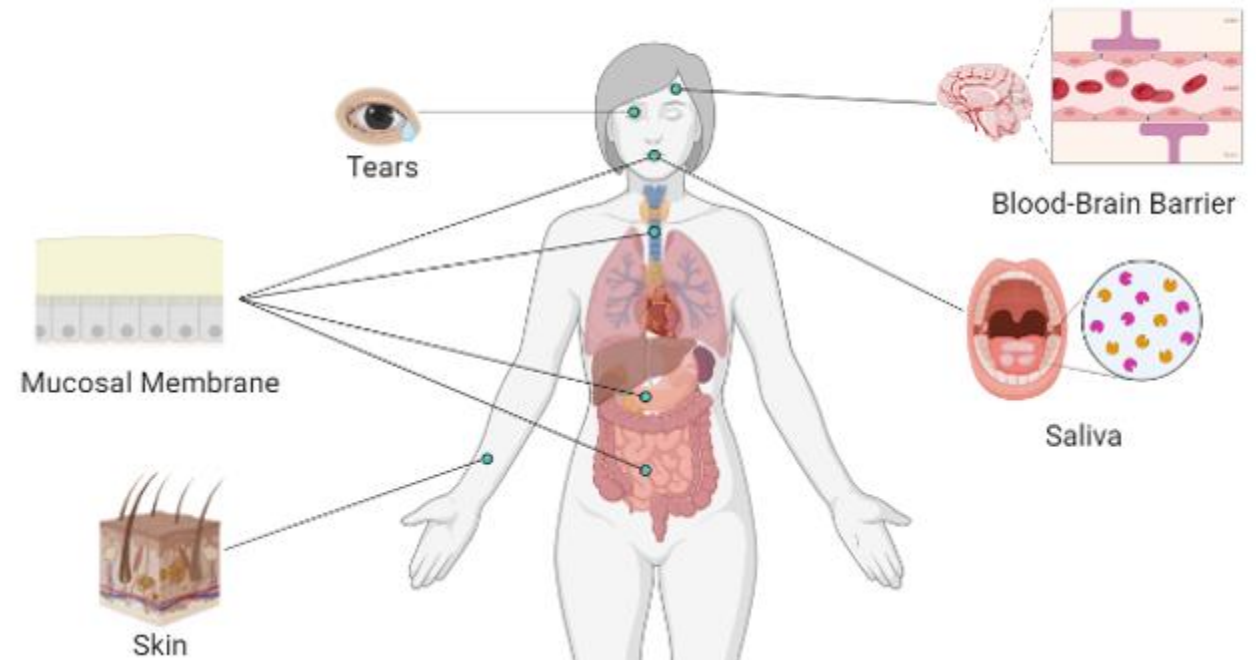
Artificial



Innate Immunity

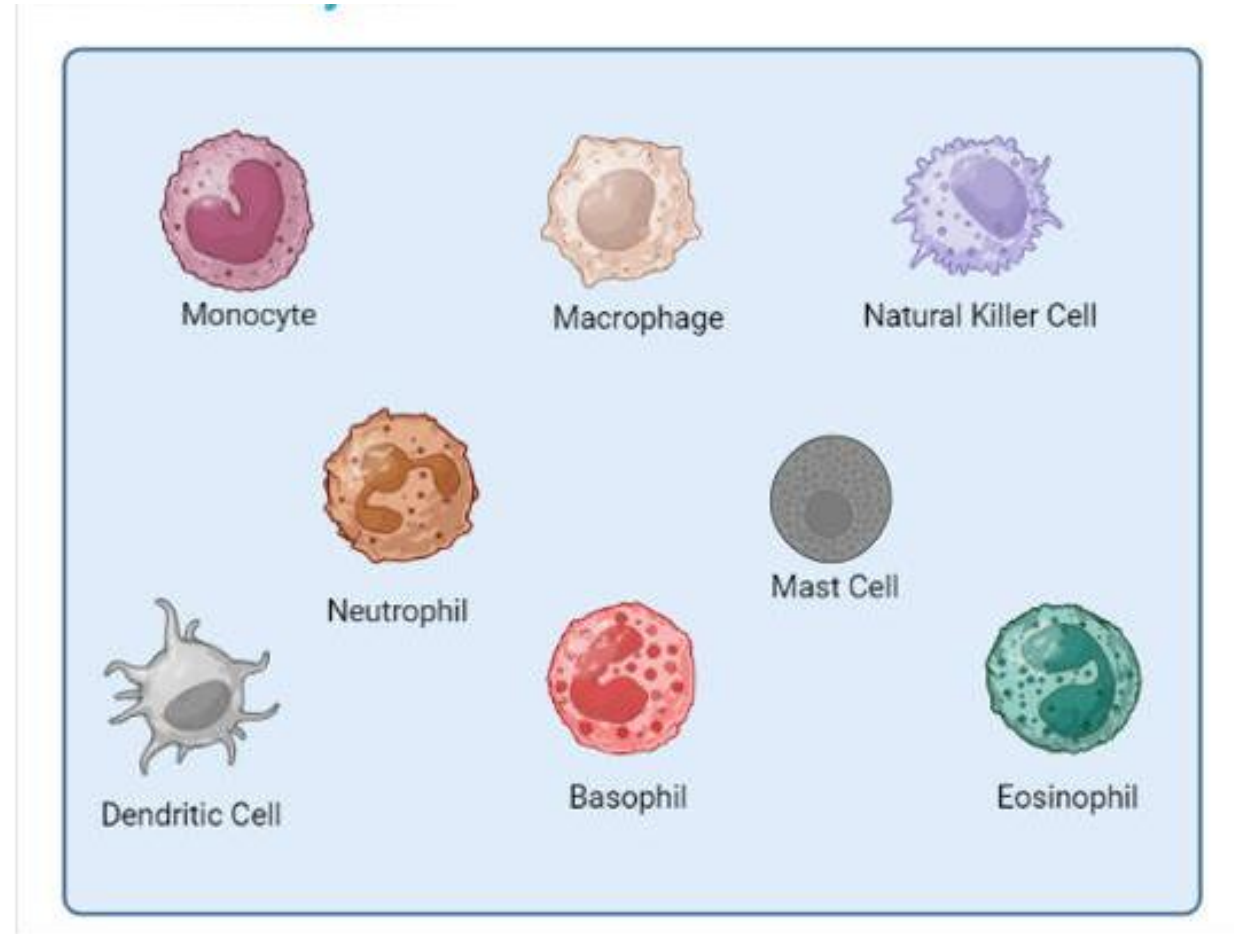
1. Innate immunity includes physical factors, chemical factors, antimicrobial proteins, natural killer cells, phagocytes, inflammation, and fever.
2. The skin and mucous membranes are the first line of defense against entry of pathogens.

| The Physical Barriers in Innate Immunity

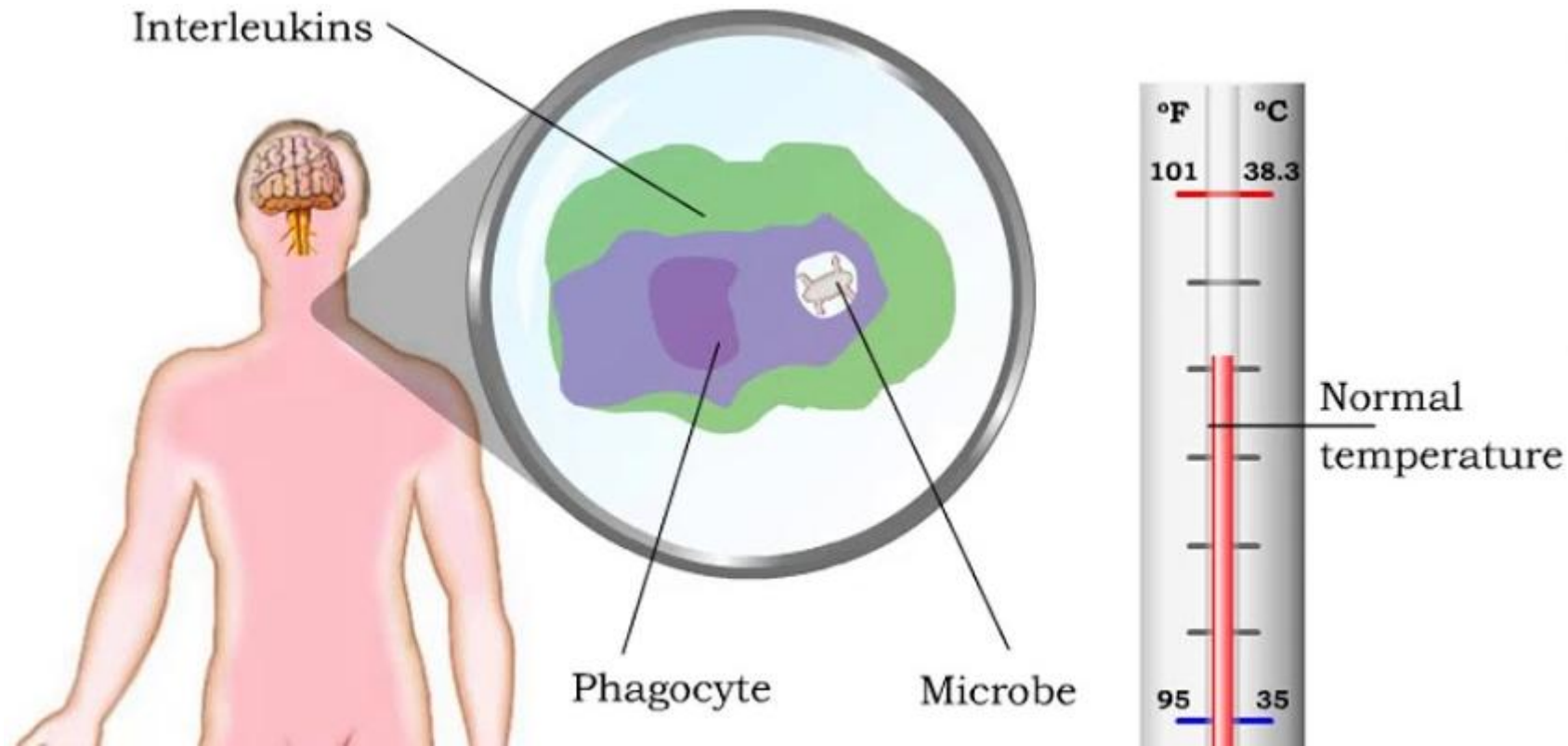


3. Antimicrobial substances include interferons, the complement system, iron-binding proteins, and anti-microbial proteins.

4. Natural killer cells and phagocytes attack and kill pathogens and defective cells in the body.

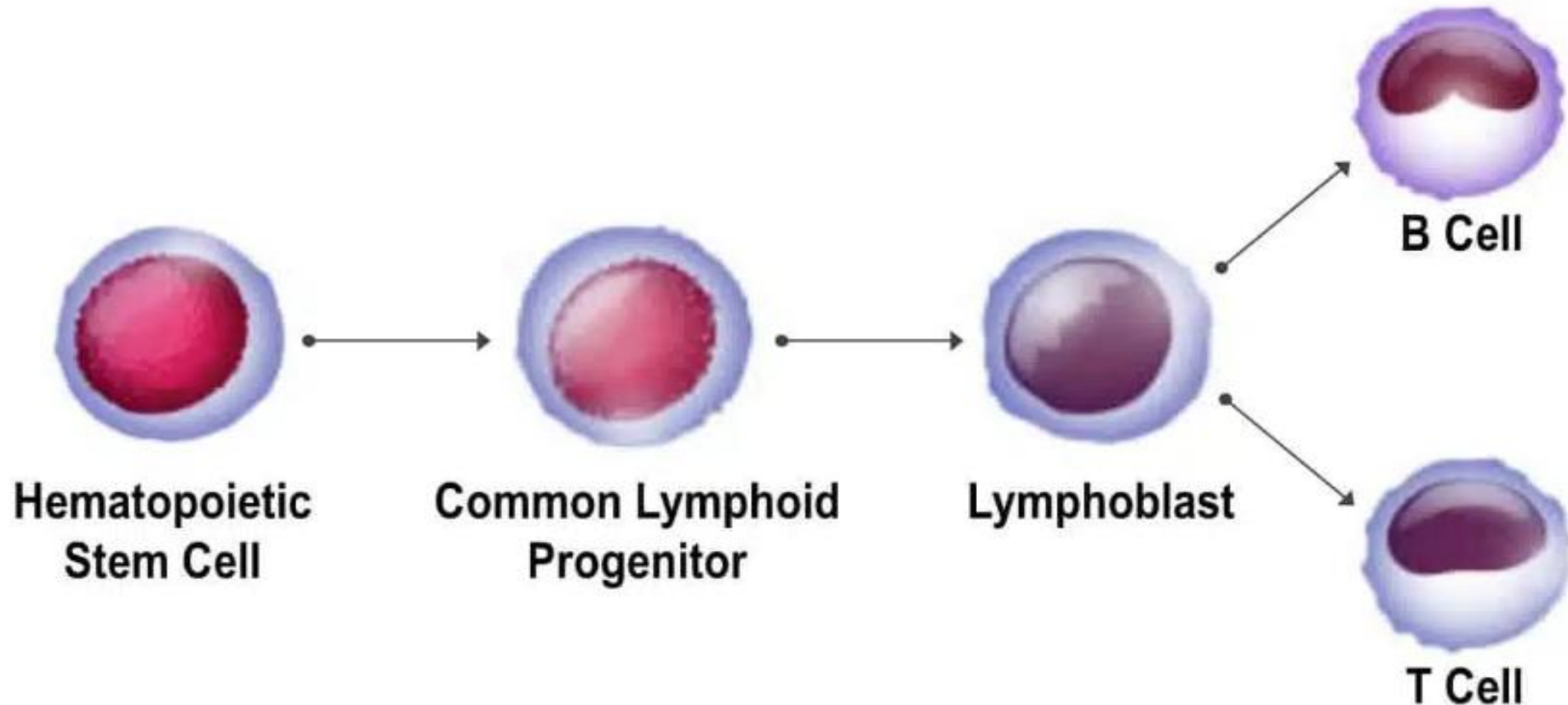


5. Inflammation aids disposal of microbes, toxins, or foreign material at the site of an injury, and prepares the site for tissue repair
6. Fever intensifies the antiviral effects of interferons, inhibits growth of some microbes, and speeds up body reactions that aid repair.

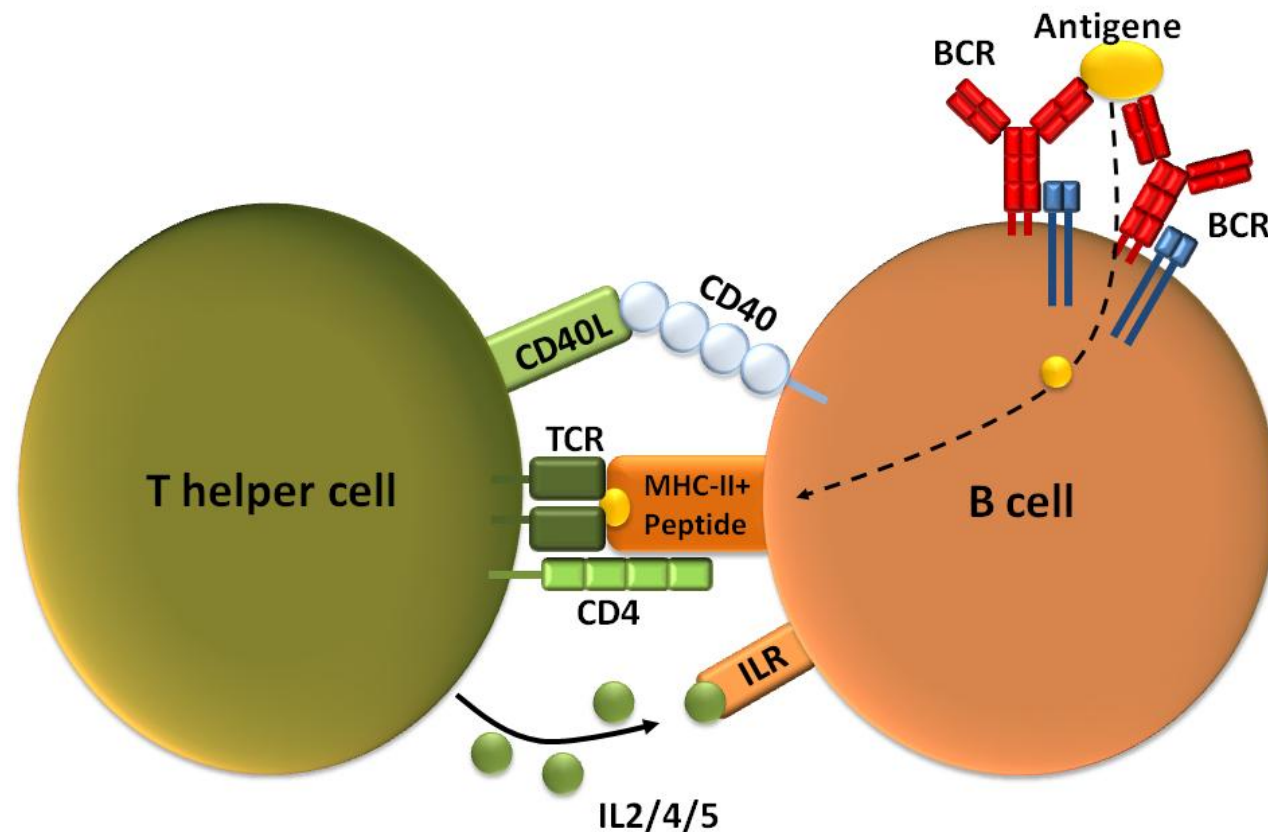


Acquired Immunity

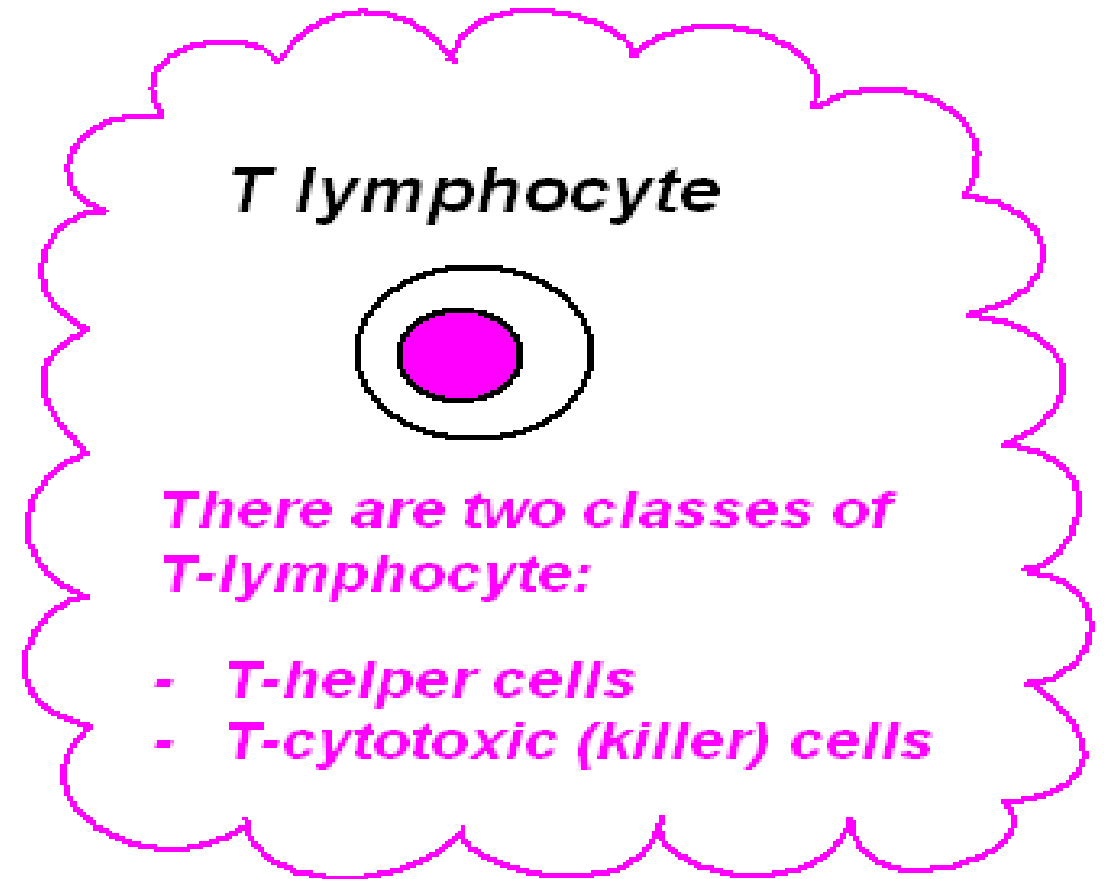
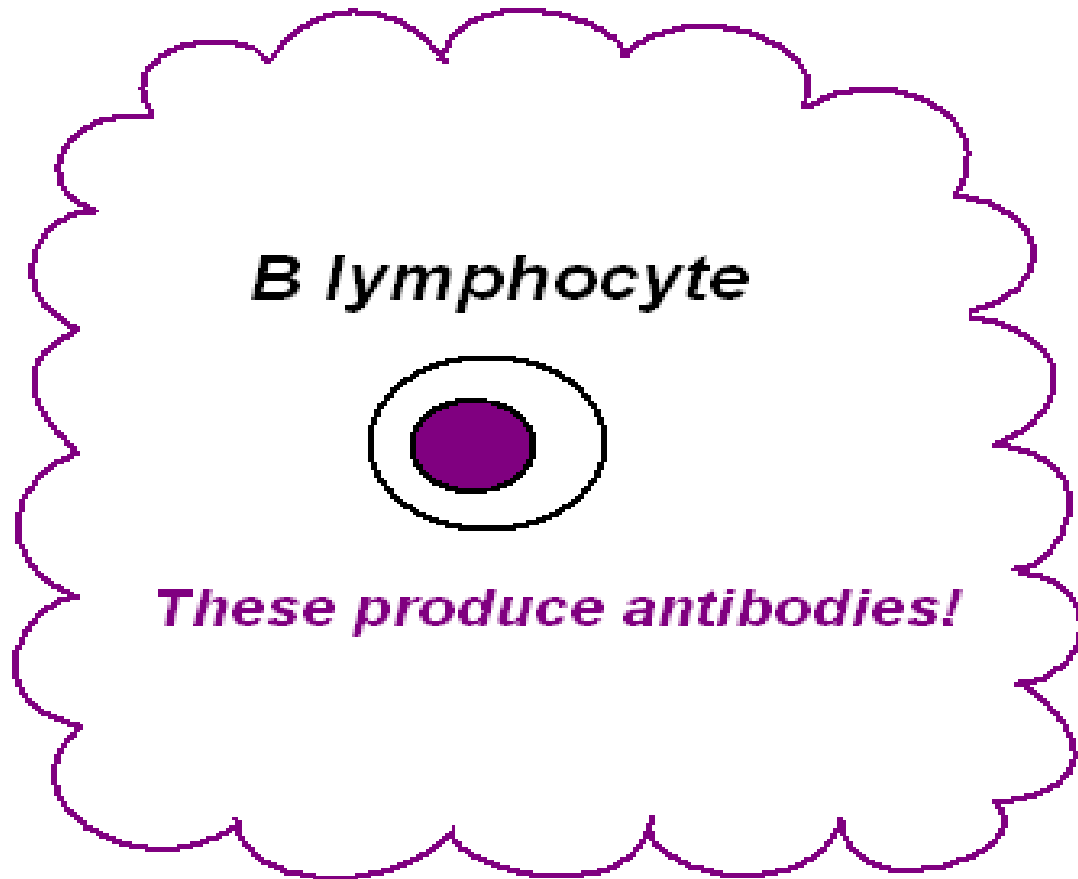
1. Acquired Immunity or Adaptive immunity involves lymphocytes called B cells and T cells. B cells and T cells arise from stem cells in red bone marrow. B cells mature in red bone marrow; T cells mature in the thymus gland.



2. Before B cells leave the red bone marrow or T cells leave the thymus, they develop immunocompetence, the ability to carry out adaptive immune responses. This process involves the insertion of antigen receptors into their plasma membranes. Antigen receptors are molecules that are capable of recognizing specific antigens.



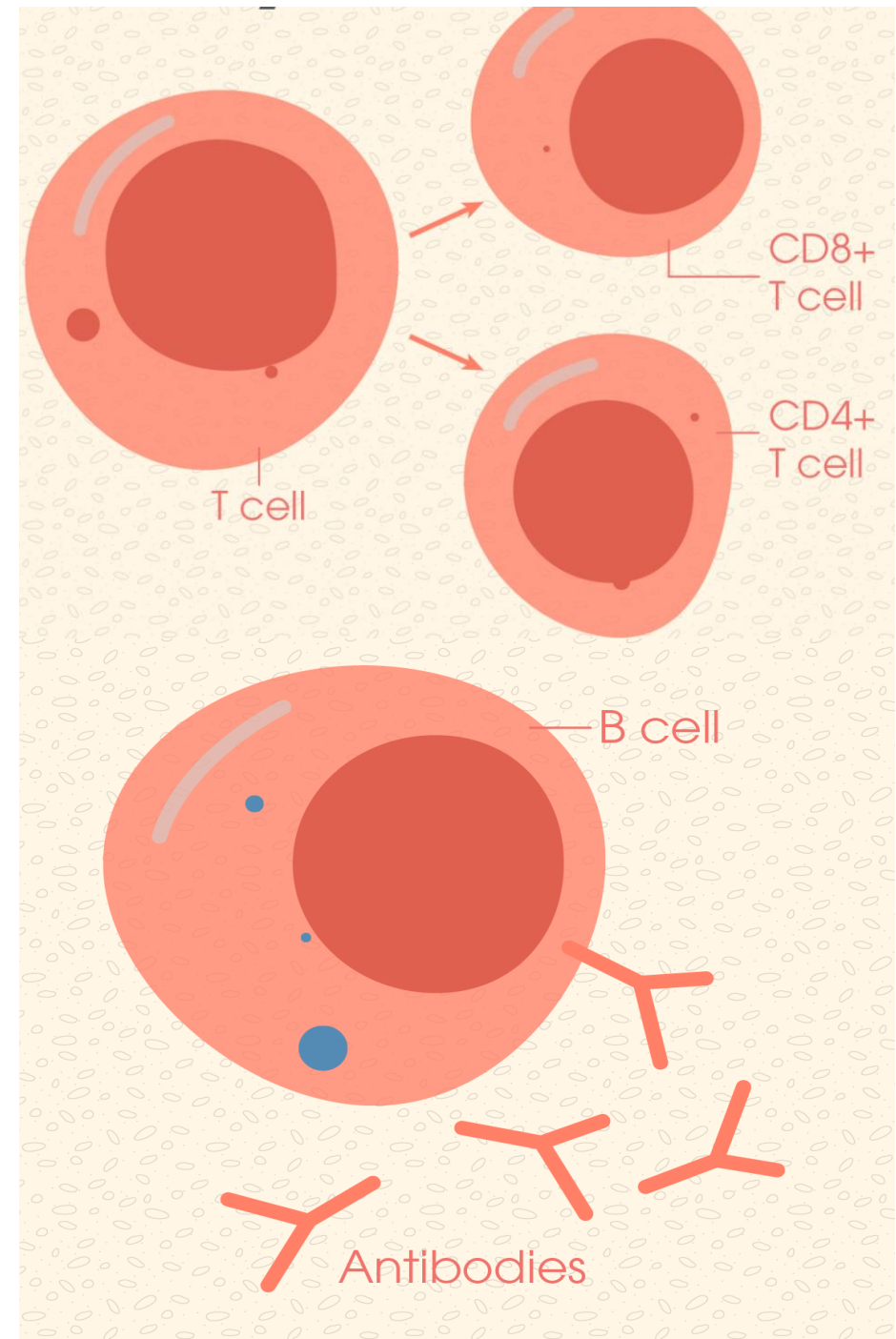
3. Two major types of mature T cells exit the thymus: helper T cells (also known as CD4 T cells) and cytotoxic T cells (also referred to as CD8 T cells).



4. There are two types of adaptive immunity: cell-mediated immunity and antibody-mediated immunity.

➤ In cell-mediated immune responses, cytotoxic T cells directly attack invading antigens.

➤ In antibody mediated immune responses B cells transform into plasma cells that secrete antibodies.



Stress and Immunity

1. Psychoneuroimmunology (PNI) deals with communication pathways that link the nervous, endocrine, and immune systems. Thoughts, feelings, moods, and beliefs influence health and the course of disease.
2. Under stress, people are less likely to eat well or exercise regularly, two habits that enhance immunity.

Aging and the Immune System

1. With advancing age, individuals become more susceptible to infections and malignancies, respond less well to vaccines, and produce more autoantibodies.
2. Immune responses also diminish with age.